Problem

Title

Sum of Digits at Prime Positions

Topics

Prime Numbers, Indexing Logic, Digit Extraction, Loops, Nested Loops, Conditional Statements

Difficulty Level

Hard

Problem Detailed Statement (Scenario Based)

In a logical reasoning class, students are given an interesting twist on digit manipulation.

The trainer provides a positive integer **N**. Students must calculate the **sum of digits that are placed at prime positions** in the number.

Important rules:

- Position indexing starts from **1 (left to right)**.
- A position is considered valid only if it is a **prime number**.
- Students must:
    1. Determine the total number of digits.
    2. Identify prime positions.
    3. Extract digits using loop-based logic.
    4. Add only those digits that lie at prime index positions.

For example:

If N = 1234567

Digits with positions:

Position	Digit
1	1
2	2
3	3
4	4
5	5
6	6

Position	Digit
7	7

Prime positions: 2, 3, 5, 7

Digits at these positions: 2, 3, 5, 7

Sum = 2 + 3 + 5 + 7 = **17**

Students are not allowed to:

- Convert number into string directly.
- Use built-in prime checking libraries.

They must:

- Use loop logic for prime checking.
- Apply nested loop where required.
- Handle edge cases such as:
  - Single digit number
  - Numbers with very few digits
  - Leading zero scenarios not applicable (input integer)

This problem strengthens:

- Prime detection logic
- Positional indexing
- Digit traversal
- Nested loop implementation
- Conditional filtering

Input Format

A single integer:

`N`

Constraints:

`1 ≤ N ≤ 10^12`

Output Format

- Print a single integer representing the sum of digits at prime positions.

Examples

Example 1

Input:

```
1234567
```

Output:

```
17
```

Example 2

Input:

```
987654
```

Positions:
Prime positions → 2, 3, 5

Digits → 8, 7, 5

Sum = 20

Output:

```
20
```

Example 3

Input:

```
5
```

(No prime positions exist)

Output:

```
0
```